

Homework: TypeScript Arrays and Loops

100 points + 5 bonus | Due before class on August 22, 2026

Student name

Date

Part 1 - Match the Vocabulary

15 points

Match: Array, Item, Index, Length, Zero-based, and Out-of-range index.

- A. One variable that stores an ordered list
- B. One value inside an array
- C. The numbered address of an item
- D. The number of items in an array
- E. A first index of 0 instead of 1
- F. An index that is not inside the array

Write the six matching letters in term order:

Explain the difference between everyday position and array index:

No computer required

Predict and explain first. You may run examples later to check your thinking.

Index and Length Practice

Homework continued

Part 2 - Index and Length Practice

20 points

```
const snacks: string[] = ["apple", "popcorn", "yogurt", "pretzels"];
```

Answer questions 1-9 with values or TypeScript lines:

Length; first index; last index; snacks[0]; snacks[2]; print pretzels; find the final item; change popcorn to carrots; explain snacks[4].

10. Explain why the length and final index are different:

Check your model

Four items have length 4. Their valid indexes are 0, 1, 2, and 3.

Predict the Output: A and B

Part 3 | 20 points total

Part 3A - Read Two Indexes

```
const levels: number[] = [2, 4, 6];  
console.log(levels[1]);  
console.log(levels.length);
```

Write the exact two output lines:

Part 3B - Trace Every Pet

```
const pets: string[] = ["cat", "dog", "fish"];  
for (let i = 0; i < pets.length; i++) {  
  console.log(`${i}: ${pets[i]}`);  
}
```

Write the exact three output lines:

Why does the first line begin with index 0?

Predict the Output: C and D

Part 3 continued

Part 3C - Add an Array

```
const numbers: number[] = [3, 5, 7];
let total: number = 0;
for (let i = 0; i < numbers.length; i++) {
  total = total + numbers[i];
}
console.log(total);
```

Show how total changes, then write the final output:

--

Part 3D - Decide for Each Temperature

```
const temperatures: number[] = [72, 85, 68];
for (let i = 0; i < temperatures.length; i++) {
  if (temperatures[i] >= 80) {
    console.log("Hot");
  } else {
    console.log("Mild");
  }
}
```

Write the exact three output lines:

--

Spot and Repair the Bugs

Part 4 | 20 points

Bug A - One Index Too Far

10 points

```
const games: string[] = ["Chess", "Soccer", "Tag"];
for (let i = 0; i <= games.length; i++) {
  console.log(games[i]);
}
```

Explain the problem and rewrite the condition:

Bug B - Accumulator in the Wrong Place

10 points

```
const points: number[] = [10, 20, 30];
for (let i = 0; i < points.length; i++) {
  let total: number = 0;
  total = total + points[i];
}
console.log(total);
```

Explain the scope problem and rewrite the complete program:

Design a List Program

Part 5 | 15 points

Part 5 - Design a List Program

15 points

Create one typed array with at least four values, loop to `.length`, read `items[i]`, and print the current item with a template literal.

Program theme and one-sentence goal:

Write the complete TypeScript program:

Describe the expected output:

AI Reflection, Bonus, and Submit

Homework conclusion

Part 6 - AI Data Reflection

10 points

```
const labels: string[] = ["tree", "unknown", "car", "tree", "unknown"];
```

Answer all five questions:

Find the length and unknown indexes. Explain why uncertainty is flagged, what a loop repeats consistently, and what decision still needs a person.

Bonus - Extend and Predict

+5 points

Add one label. Give the new array, length, final index, and expected output for the added item.

Submit to Microsoft Teams AND Ishwari Raut ma'am.